

10 Moment hybnosti

$$\vec{L} = \vec{r} \times \vec{p}$$

$$[L_j^A, L_k^A] = i\hbar \sum_l \epsilon_{jkl} L_l^A$$

$$L^2 = L_x^2 + L_y^2 + L_z^2 \quad ; \quad [L^2, L_j^A] = 0$$

- předpokládáme, že máme řešení L^2 a L_z^A :

3.1 Kvantování momentu hybnosti:

→ ústředí problém ⇒ máme 2 kvantová čísla λ, m

$$L^2 |\lambda, m\rangle = \lambda \hbar^2 |\lambda, m\rangle \quad L_z^A |\lambda, m\rangle = m \hbar |\lambda, m\rangle$$

1. přípravný bod

⇒ analogicky anihilující a kvantující operátory:

$$L_+^A := L_x^A + iL_y^A \quad L_-^A := L_x^A - iL_y^A$$

$$L_+^A = L_-^A \quad L_-^A = L_+^A$$

$$[L_z^A, L_-^A] = [L_z^A, L_x^A] - i[L_z^A, L_y^A] =$$

$$= L_y^A i\hbar - i(-i\hbar L_x^A) = -\hbar L_-^A$$

$$[L_z^A, L_+^A] = \hbar L_+^A$$

$$[L^2, L_\pm^A] = 0$$

$$L_-^A L_+^A = (L_x^A - iL_y^A)(L_x^A + iL_y^A) = L_x^2 + L_y^2 + i(L_x^A L_y^A - L_y^A L_x^A)$$

$$= L_x^2 + L_y^2 + i(\hbar)L_z^A = L^2 - L_z^2 - \hbar L_z^A$$

$$L_+^A L_-^A = L^2 - L_z^2 + \hbar L_z^A$$

2. přípravný bod

$$L_\pm^A (L_\pm^A |\lambda, m\rangle) = L_\pm^A (L^2 |\lambda, m\rangle) = L_\pm^A (\lambda \hbar^2 |\lambda, m\rangle)$$

$$= \lambda \hbar^2 L_\pm^A |\lambda, m\rangle$$

$$L_z^A (L_+^A |\lambda, m\rangle) = (\hbar L_+^A + L_+^A L_z^A) |\lambda, m\rangle$$

$$= \hbar(m+1) L_+^A |\lambda, m\rangle$$

$$L_z^A (L_-^A |\lambda, m\rangle) = \hbar(m-1) L_-^A |\lambda, m\rangle$$

3. přípravný bod

4. vzhled:

$$\langle 2 | L_-^A L_+^A | 2 \rangle = \langle L_+^A 2 | L_+^A 2 \rangle = \|L_+^A 2\|^2 \geq 0$$

$$\langle 2 | L_+^A L_-^A | 2 \rangle \geq 0 \quad \text{analogicky}$$

$$\frac{1}{2} (L_+^A L_-^A + L_-^A L_+^A) = L^2 - L_z^2$$

$$2 = |\lambda, m\rangle \quad 0 \leq \langle \lambda, m | \frac{1}{2} (L_+^A L_-^A + L_-^A L_+^A) | \lambda, m \rangle$$

$$= \langle \lambda, m | L^2 - L_z^2 | \lambda, m \rangle = \lambda \hbar^2 - m^2 \hbar^2$$

$$\Rightarrow \lambda \geq m^2 \quad \wedge \quad \lambda \geq 0$$

$$\sqrt{\lambda} \geq |m|$$

$$\lambda \text{ celé} \Rightarrow \lambda = m \cdot \{m\} \quad \lambda' = m \cdot \{m\}$$

$$\rightarrow \lambda \geq \lambda' \quad , \quad |\lambda, \lambda\rangle \neq 0 \quad \wedge \quad L_+^A |\lambda, \lambda\rangle = 0$$

$$|\lambda, \lambda'\rangle \neq 0 \quad \wedge \quad L_-^A |\lambda, \lambda'\rangle = 0$$

~ $|\lambda, \lambda+1\rangle$
2+1 by bylo větší vlastně číslo
š s tím, že
2 je největší

4. přípravný bod

$$0 = L_-^A L_+^A |\lambda, \lambda\rangle = (L^2 - L_z^2 - \hbar L_z^A) |\lambda, \lambda\rangle$$

$$= (\lambda \hbar^2 - \lambda^2 \hbar^2 - \lambda \hbar^2) |\lambda, \lambda\rangle$$

$$0 = L_+^A L_-^A |\lambda, \lambda'\rangle = (L^2 - L_z^2 + \hbar L_z^A) |\lambda, \lambda'\rangle$$

$$= (\lambda \hbar^2 - \lambda'^2 \hbar^2 + \lambda' \hbar^2) |\lambda, \lambda'\rangle$$

$$\Rightarrow \lambda - \lambda'^2 - \lambda = 0$$

$$\Rightarrow \lambda' = \lambda + 1 \quad \text{společně}$$

$$\lambda - \lambda'^2 + \lambda' = 0$$

$$\lambda' = -\lambda$$

$$\lambda = \lambda(\lambda+1)$$

$$\Rightarrow |\lambda, m\rangle \rightarrow |\lambda, m\rangle$$

5. přípravný bod

$$L_-^A |\lambda, \lambda\rangle \sim |\lambda, \lambda-1\rangle$$

$$L_-^A |\lambda, \lambda\rangle \sim |\lambda, \lambda-s\rangle$$

$$\rightarrow \lambda-s = -\lambda \Rightarrow s = 2\lambda \in \mathbb{N}_0$$

$$\Rightarrow \lambda = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$$

$$L^2 \quad \lambda(\lambda+1)\hbar^2 \quad \lambda = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$$

$$L_z^A \quad 2\lambda+1 \quad \text{hodnot: } m = \lambda, \lambda-1, \dots, -\lambda+1, -\lambda$$

obecně:

$$L_+^A |\lambda, m\rangle = c_{\lambda m}^+ |\lambda, m+1\rangle$$

$$L_-^A |\lambda, m\rangle = c_{\lambda m}^- |\lambda, m-1\rangle$$

$$\langle \lambda, m | L_-^A L_+^A | \lambda, m \rangle = \|L_+^A |\lambda, m\rangle\|^2 = |c_{\lambda m}^+|^2$$

$$\langle \lambda, m | L^2 - L_z^2 - \hbar L_z^A | \lambda, m \rangle = \hbar^2 [\lambda(\lambda+1) - m^2 - m]$$

$$\text{BÚNO ai m fci: } c_{\lambda m}^+ = \hbar \sqrt{\lambda(\lambda+1) - m^2 - m}$$

$$= \hbar \sqrt{(l-m)(l+m+1)}$$

anal.:

$$c_{\lambda m}^- = \hbar \sqrt{\lambda(\lambda+1) - m^2 + m}$$

$$= \hbar \sqrt{(l+m)(l-m+1)}$$

3.2 Maticová reprezentace

$$\beta = \{ |\lambda, m\rangle : \lambda = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots ; m = \lambda, \dots, -\lambda \}$$

$$(L^2)_\beta = \hbar^2 \begin{pmatrix} 0 & & & 0 \\ \frac{3}{4} & & & \\ & \frac{1}{4} & & \\ 0 & & \ddots & \ddots \end{pmatrix}$$

$$(L_z^A)_\beta = \begin{pmatrix} 0 & \frac{1}{2} & & 0 \\ & \frac{1}{2} & & \\ 0 & & \ddots & \\ & & & -\frac{1}{2} \end{pmatrix} \hbar$$

$$\langle \lambda', m' | L_+^A | \lambda, m \rangle = \langle \lambda', m' | c_{\lambda m}^+ | \lambda, m+1 \rangle$$

$$= c_{\lambda m}^+ \delta_{\lambda \lambda'} \delta_{(m+1)m'}$$

$$L_x^A = \frac{1}{2} (L_+^A + L_-^A) \quad L_y^A = \frac{1}{2i} (L_+^A - L_-^A)$$

$$|\lambda, m\rangle : |\lambda, \lambda\rangle, |\lambda, \lambda-1\rangle, \dots, |\lambda, -\lambda\rangle$$

$$\lambda=0 \quad \|L_z\| = \|L_+\| = \|L_-\| = \|L_x\| = \|L_y\| = \|L^2\| = 0$$

$$\lambda = \frac{1}{2} \rightarrow \left| \frac{1}{2}, \frac{1}{2} \right\rangle, \left| \frac{1}{2}, -\frac{1}{2} \right\rangle$$

$$\Rightarrow c_{\frac{1}{2} \frac{1}{2}}^+ = 0 \quad c_{\frac{1}{2} -\frac{1}{2}}^+ = \hbar$$

$$\left\langle \frac{1}{2}, \frac{1}{2} \left| L_+^A \right| \frac{1}{2}, \frac{1}{2} \right\rangle = 0$$

$$\left\langle \frac{1}{2}, \frac{1}{2} \left| L_+^A \right| \frac{1}{2}, -\frac{1}{2} \right\rangle = \hbar$$

$$\left\langle \frac{1}{2}, -\frac{1}{2} \left| L_+^A \right| \frac{1}{2}, \frac{1}{2} \right\rangle = 0$$

$$\left\langle -\frac{1}{2}, -\frac{1}{2} \left| L_+^A \right| -\frac{1}{2}, -\frac{1}{2} \right\rangle = 0$$

$$\Rightarrow \|L_-\| \text{ sdružený operátor k } \|L_+\| \Rightarrow \|L_-\| = \hbar \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$$\Rightarrow \|L_x\| = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \|L_y\| = \frac{\hbar}{2i} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\|L_z\| = \hbar \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

→ Pauliho matice: $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ σ_z je diagonální

$$\Rightarrow \|L_x\| = \frac{\hbar}{2} \sigma_x \quad \|L_y\| = \frac{\hbar}{2} \sigma_y \quad \|L_z\| = \frac{\hbar}{2} \sigma_z$$

10.3 Vlastní funkce operátoru orbitálního L

bylo asi už bylo

$$L^2 |\lambda, m\rangle = \lambda(\lambda+1) \hbar^2 |\lambda, m\rangle \quad L_z |\lambda, m\rangle = m \hbar |\lambda, m\rangle$$

$$|\lambda, m\rangle = Y_{\lambda m}(\theta, \varphi) = N_{\lambda m} P_{\lambda}^{(m)}(\cos \theta) e^{im\varphi}$$

$$N_{\lambda m} = \sqrt{\frac{(l-m)!}{(l+m)!} \frac{(2l+1)}{4\pi}}$$

$$P_{\lambda}^{(m)}$$
 je polynom Legendreho

$$P_{\lambda}(\xi) = \frac{1}{2^{\lambda} \lambda!} \frac{d^{\lambda}}{d\xi^{\lambda}} (\xi^2-1)^{\lambda}$$

$$P_{\lambda}^{(m)}(\xi) = (1-\xi^2)^{\frac{m}{2}} \frac{d^{l-m}}{d\xi^{l-m}} P_{\lambda}(\xi)$$

$$P_{\lambda}^0(\xi) = P_{\lambda}(\xi)$$

$$P_0(\xi) = 1$$

$$P_1(\xi) = \xi$$

$$P_1^1(\xi) = \sqrt{1-\xi^2}$$

$$Y_0^0(\theta, \varphi) = \frac{1}{\sqrt{4\pi}}$$

$$Y_1^0(\theta, \varphi) = \sqrt{\frac{3}{4\pi}} \cos \theta$$

$$Y_1^1(\theta, \varphi) = \sqrt{\frac{3}{8\pi}} \sin \theta e^{i\varphi}$$

$$Y_1^{-1}(\theta, \varphi) = \sqrt{\frac{3}{8\pi}} \sin \theta e^{-i\varphi}$$